Python Practice: Dictionaries

Part 4 of 5 | 10 questions | Beginner to Expert

RULES: No for/while loops, no comprehensions, no recursion, no map/filter/reduce, and no hardcoded answers. Use indexing, slicing, operators, built-in functions, and data structure methods.

LEVEL: BEGINNER

Q31. Dictionary Access

```
student = {"name": "Ali", "age": 22, "course": "Python", "marks": 85}
```

Extract name and marks; retrieve missing city with default Karachi.

Q32. Update Dictionary

```
employee = {"name": "Sara", "salary": 100000, "department": "IT"}
```

Raise salary by 20%, add bonus 15000, change department to Data Engineering.

Q33. Dictionary Methods

```
product = {"id": 101, "name": "Laptop", "price": 120000, "stock": 5}
```

Extract keys, values, items; remove stock and retain its value.

LEVEL: INTERMEDIATE

Q34. Merge Dictionaries

```
dict1 = {"a": 10, "b": 20, "c": 30}
dict2 = {"b": 50, "d": 40}
```

Merge so dict2 overrides overlapping keys.

Q35. Student Marks

```
marks = {"Ali": 85, "Sara": 92, "Ahmed": 78, "Zain": 88}
```

Find highest, lowest, average, and top-scoring student without loops or lambda.

Q36. Nested Employee Records

```
employees = {
    "E101": {"name": "Ali", "salary": 80000},
    "E102": {"name": "Sara", "salary": 120000},
    "E103": {"name": "Ahmed", "salary": 95000}
}
```

Extract Sara salary, increase Ali salary 15%, and calculate total payroll.

Q37. Dictionary Inversion

```
data = {"Python": "Ali", "SQL": "Sara", "AWS": "Ahmed"}
```

Create inverse dictionary without loops or comprehensions.

LEVEL: ADVANCED

Q38. Order Calculation

```
order = {
    "laptop": {"price": 120000, "quantity": 2},
    "mouse": {"price": 1500, "quantity": 5},
    "keyboard": {"price": 3500, "quantity": 3}
}
```

Calculate subtotals, grand total, and amount after 10% discount.

Q39. Employee Performance

```
employees = {
    "Ali": {"python": 90, "sql": 85, "aws": 88},
    "Sara": {"python": 95, "sql": 92, "aws": 90},
    "Ahmed": {"python": 80, "sql": 87, "aws": 84}
}
```

Calculate each average, top performer, and highest SQL score.

Q40. Inventory Management

```
inventory = {
    "P101": {"name": "Laptop", "price": 120000, "stock": 5},
    "P102": {"name": "Mouse", "price": 1500, "stock": 20},
    "P103": {"name": "Keyboard", "price": 3500, "stock": 10}
}
```

Calculate total value, most valuable item by price x stock, and update stock after selling two laptops and five mice.